

Antifouling Slippery Surface against Marine Biofouling

Xin Liu, Xiaolei Gu, Yuyang Zhou, Weihao Pan, Jiyu Liu, and Jinlong Song*



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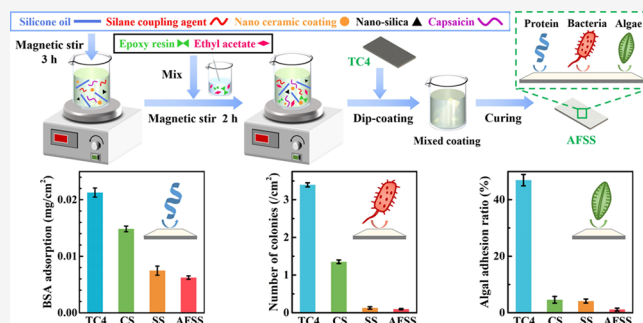
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ABSTRACT: Titanium and its alloys have become the most excellent structure materials for naval seawater pipelines due to their high strength and good corrosion resistance. However, marine biofouling poses a serious threat to titanium alloy piping systems because of their good biocompatibility. Recently, the biomimetic antifouling coating, a novel antifouling method, has received great attention. Here, based on this biomimetic idea, we develop a nontoxic antifouling slippery surface (AFSS) using silicone oil, silane coupling agent, nanosilica, nanoceramic coating, epoxy resin, and capsaicin. The developed AFSS has excellent slippery performance for various droplets, good durability, and a superior self-cleaning property. Additionally, the antifouling performance of the AFSS was significantly enhanced, as confirmed by the reduced adhesion of proteins (70.7%), bacteria (97.2%), and algae (97.7%) compared to the ordinary titanium alloy. With these excellent properties, the AFSS was expected to be a promising candidate for protecting titanium alloy piping systems from marine biofouling.



INTRODUCTION

Titanium and titanium alloys have become the most excellent structural materials for naval seawater pipelines because of their properties such as high specific strength, low density, corrosion resistance, nonmagnetic properties, good weldability, and great impact resistance.¹ However, due to the great biocompatibility of titanium, biofouling is the main problem of the titanium alloy piping systems. Marine organisms, including microscopic bacteria and algae, enter the seawater pipeline through the submarine valve boxes and then settle and propagate onto the inner wall. The attached biological communities reduce the diameter of the pipeline, hinder the opening and closing of the valves, and even clog the pipelines. These pipeline tumors seriously endanger the normal running of the seawater pipeline system² and indirectly lead to reduced navigation speed, environmental pollution, and economic losses. Therefore, there is an urgent need to prevent marine biofouling in the titanium alloy pipeline systems.

To date, various methods have been proposed to prevent marine biofouling, such as physical methods, chemical methods, and biomimetic methods. Physical antifouling methods, including ultrasound,^{3,4} microbubbles blasting,⁵ water jet impact,⁶ ultraviolet radiation,⁶ and high-temperature inactivation,⁶ have the disadvantages of complex technologies and high cost. Chemical antifouling methods, such as seawater electrolysis⁷ and coating protection,⁸ aim to kill the spores or larvae of marine organisms attached to the pipelines by selecting effective chemicals. However, these methods have great toxicity to both fouling organisms and nontarget marine organisms, thus endangering the ecological balance of marine

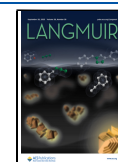
systems. Accordingly, it is of great importance to develop an effective and environmentally friendly antifouling method applicable to the titanium alloy pipeline systems. Inspired by the antifouling phenomenon of animals and plants in nature, biomimetic antifouling methods have been proposed to prevent the attachment of marine organisms.

As a novel antifouling method, biomimetic antifouling coating surfaces have received a lot of attention in recent years. There are two kinds of biomimetic antifouling coating surfaces. One is to extract the secretions from organisms with antifouling properties and add the secretions to coatings as antifouling agents. Natural antifouling agents are mainly found in marine invertebrates, marine plants, marine microorganisms, and terrestrial plants. Zhang et al. evaluated the antibacterial performance of 12 compounds in marine mollusk coral, but none showed broad-spectrum antifouling performance.⁹ Nys et al. studied the antifouling performance of bromofuranone,¹⁰ halogenated sesquiterpene type, and sesquiterpene acetylene esters extracted from marine plant red algae, which also lacked the broad-spectrum antifouling performance.¹¹ Desbois et al. found that the extracts of marine microorganism diatom had strong antibacterial activity against *Staphylococcus aureus*.¹² Moreover, capsaicin and its synthetic analogues, an extract of

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pepper, have attracted extensive attention due to its excellent antifouling property and green nontoxic property.^{13–16} Peng et al. found that coatings with a capsaicin content of 1–5% showed a good antifouling property in the short term, but the antifouling performance was reduced due to the loss of capsaicin in aqueous solution.¹⁷

Another biomimetic antifouling method is to construct surfaces with special wettability by mimicking the structures and functions of animals and plants in nature, viz., superhydrophobic surfaces¹⁸ and slippery surfaces.¹⁹ Chien et al. found that the prepared shark skin replicas could effectively prevent the formation of a biofilm.²⁰ However, the research results suggested that microorganisms, such as bacteria or diatoms, can gradually fill in the microtextured surfaces, thus destroying the antifouling performance.²¹ Furthermore, the air trapped within surface profiles cannot withstand high pressure, which could be easily displaced by water ultimately, thus leading to the loss of the antifouling performance.²² Recently, slippery liquid-infused porous surfaces (SLIPS) have been introduced as an antifouling alternative approach²³ to superhydrophobic surfaces. It has been shown that SLIPS have many other excellent properties, such as anticorrosion,^{24,25} anti-icing,^{26,27} self-cleaning,^{28,29} water mist capture,^{30,31} and efficient condensation.³² Epstein et al. reported SLIPS constructed on polytetrafluoroethylene (PTFE) films³³ and evaluated their excellent antibacterial capability. Ouyang et al. reported an oil-infused surface on a stainless steel substrate³⁴ and the density of settled marine bacteria on the SLIPS was two orders of magnitude lower than that on bare stainless steel. However, the antifouling performance of SLIPS is primarily conferred by the infused lubricant which has difficulty in tightly binding to the substrate.³⁵ In addition, the shear stress caused by flowing water accelerates the loss of lubricant, thus shortening the service life of the SLIPS. Slippery coatings with high substrate adhesion strength could overcome this problem. However, the existing slippery coatings have disadvantages including low slippery performance, harsh preparation conditions, utilization of toxic chemical reagents, and unreported antifouling performance against proteins, bacteria, and algae.

In this paper, we investigated the preparation of slippery surfaces without free lubricant, which combined two biomimetic antifouling methods to develop an antifouling slippery surface (AFSS) using capsaicin, silicone oil, silane coupling agent, nanosilica, nanoceramic coating, and epoxy resin. Compared with other biomimetic antifouling methods, we prepared a slippery surface with a simple process, lower cost, and better universality. The addition of nanoparticles effectively improved the mechanical strength of the coating and enhanced its durability, and the addition of capsaicin had green antifouling activity and enhanced the antifouling ability of the surface. The as-prepared surface with low surface energy exhibited good slippery performance for various droplets and had a good self-cleaning property as well as water immersion durability. Additionally, the AFSS possessed excellent anti-protein, antibacterial, and antialgal adhesion properties, which may have promising application prospects in the marine antifouling aspect of titanium alloys.

EXPERIMENTAL SECTION

Materials. Commercially available titanium alloy plates (Ti-6Al-4 V#, TC4) were purchased from Wenghou Metal Material Co., Ltd. (China). GF-106 nanoceramic coating (NCC) was purchased from

Dongguan Guangfu New Material Co., Ltd. (China). Dimethyl silicone oil (SO) was obtained from Dow Corning Co., Ltd. (USA). The density and surface tension of the SO at room temperature were ~ 0.96 g/cm³ and ~ 20.5 mN/m, respectively. Silane coupling agent (SCA) and epoxy resin (ER) were purchased from Shandong Yusuo Chemical Technology Co., Ltd. (China). Ethyl acetate (EA) was purchased from Tianda Chemical Reagent Factory (China). Nano-silica (NS) was purchased from Shanghai Yaoyi Alloy Material Co., Ltd. (China). Capsaicin was obtained from Hefei Kenu Biological Technology Co., Ltd. (China). Bovine serum protein (BSA) used for the antiprotein test was purchased from Hefei Qiansheng Biotechnology (China). *E. coli* and *S. aureus* were purchased from Shanghai Luwei Strain Biotechnology Research Center (China). Concentrated algal solution used for the antialgal test was purchased from Shandong Binzhou Fengyun algae industry (China).

Characterization. The contact angle (CA) was measured using an optical contact angle meter (Solon, SL200KS, China) by dropping a water droplet onto the surfaces under the condition of room temperature. The sliding angle (SA) was measured by using a water droplet to determine the slippery performance of the surfaces. Surface morphology was observed by a superhigh magnification lens zoom 3D microscope (Keenes, VHX-600E, Japan). The surface composition was analyzed by energy-dispersive spectroscopy (EDS, INCAs Energy, Oxford Instruments, UK) and X-ray photoelectron spectroscopy (XPS, Thermo, ESCALAB 250Xi, USA). The hardness of the AFSS was measured by a QHQ-A pencil scratch hardness tester. The adhesion was measured by the SISI adhesion guide grid tester. Photos and videos of the surfaces were taken by a digital camera (Canon, DS00, Japan).

Preparation of the AFSS. Figure 1 shows the preparation process of the AFSS. Before preparing the coating solution, the titanium alloy

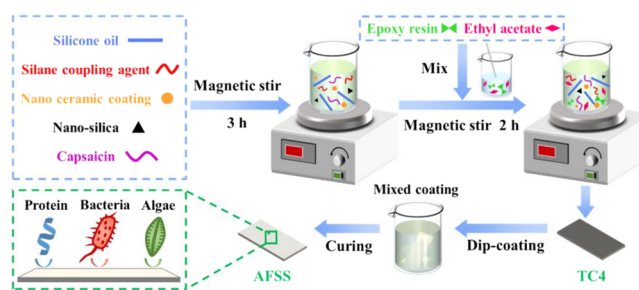


Figure 1. Schematic of the preparation processes of the AFSS.

plates were ultrasonically cleaned with anhydrous ethanol and dried in the oven to remove impurities and dirt on the surface. At room temperature, NCC, SO, SCA, NS, and capsaicin were mixed in a beaker in a certain ratio and magnetic stirred for 3 h at 600 r/min. ER was added to ethyl acetate and stirred until completely dissolved, then added to the above solution, and magnetically stirred at 600 r/min for 2 h. The pretreated titanium alloy plates were dipped into the prepared coating solution, then taken out, and dried at room temperature for 48 h to make the coating solution curing.

Self-Cleaning Test. For the self-cleaning test, the samples were fixed on a stage with a tilted angle of 30°. The hydrophilic inorganic contaminant used in the test was copper chloride dihydrate, and the hydrophobic inorganic contaminant was nanosilica particles. A mass of 0.1 g of the hydrophilic and hydrophobic inorganic contaminants was sprinkled onto the surfaces of the samples. The deionized water with a volume of 4 mL was dropped evenly on the top of the samples to test the self-cleaning performance of the samples.

Antiprotein Test. For the antiprotein test, the samples were immersed in phosphate buffer (0.1 M, pH = 7.0), washed, and dried to remove dirt and protein. The protein solution with a concentration of 1 mg/mL was prepared by dissolving 200 mg of bovine serum protein (BSA) in 200 mL of phosphate buffer (0.1 M, pH = 7.0). The standard protein solution was diluted, and the A280 value of each concentration was measured at 280 nm. The standard curve (Figure

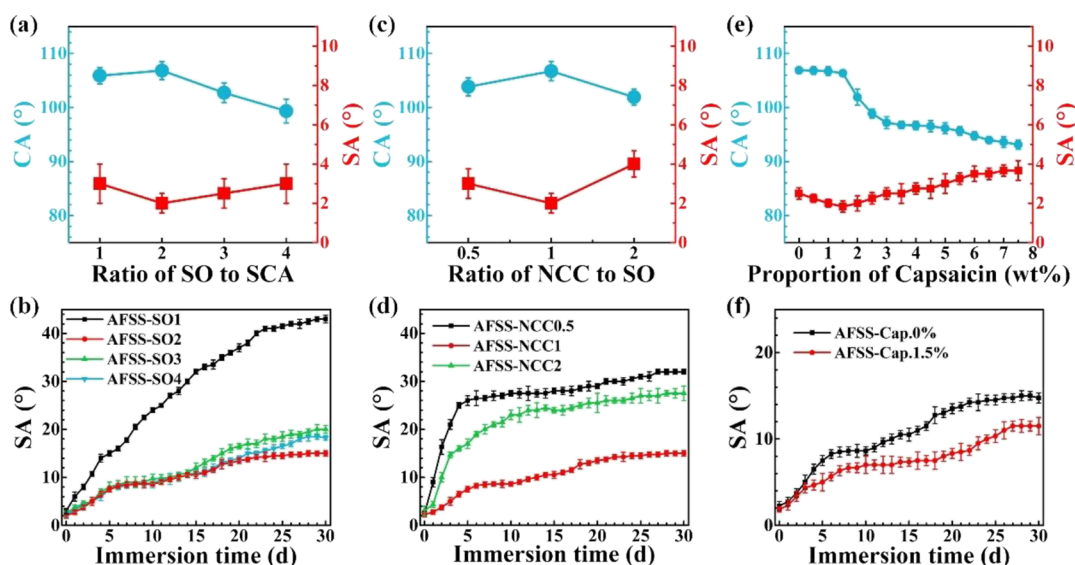


Figure 2. Influence of the concentration of different components on the CA and SA of the AFSS: (a) SO, (c) NCC, and (e) capsaicin. Variation of SA with immersion days: (b) different ratios of SO, (d) different ratios of NCC, and (f) 0 and 1.5 wt % capsaicin.

S1) was plotted using the A280 value as the vertical coordinate and the protein concentration as the horizontal coordinate. The samples were immersed in 10 mL of the standard protein solution and stored for 12 h at room temperature to reach adsorption–desorption equilibrium. After adsorption, the samples were taken out, and the concentration of the remaining protein solution was determined using an ultraviolet–visible spectrophotometer (UV–vis, INESA Analytical Instrument, N4S, China) to calculate the amount of contaminants adsorbed on the membrane surface.

Antibacterial Test. For the antibacterial test, *E. coli* and *S. aureus* were selected as the test strains. The samples, beakers, and medium were first sterilized in a vertical pressure steam sterilizer at 121 °C for 1 h. The initial bacterial solution was diluted five times to reach 10^{-5} times the original concentration. The sterilized samples were placed vertically in 10 mL of the diluted bacterial solution and incubated in a bioincubator at 37 °C for 24 h. After incubation, the samples were taken out from the bacterial solution, rinsed with distilled water, and dried. The surface colonies were observed by plate counting. The experimental process of the antibacterial test is shown in Figure S2.

Antialgal Test. For the antialgal test, the benthic diatoms were selected as the test species. The samples, beakers, and clear glassware were cleaned and dried. The diluted algal solution was prepared by artificial seawater and concentrated algal solution at a mass mixing ratio of 3:1. The algal solution was prepared by mixing the nutrient salts and the diluted algal solution at a ratio of 1:1000 according to the *F/2* formula.³⁶ The mixed algal solution was incubated in a bioincubator at 25 °C for 48 h. The light conditions for the test were set to natural light, with a controlled diurnal light ratio of 12 h:12 h. The bottles were shaken gently for 5 min daily to dissolve carbon dioxide and promote the growth and reproduction of the algae. The samples were immersed vertically in the resulting algal solution and incubated in a bio-incubator with a temperature of 25 °C for 7 days. After incubation, the samples were taken out from the algal solution, rinsed with distilled water, and dried. The surfaces of the samples were observed using a fluorescent inverted microscope (FIM, Oplonic Optonics, M-FL-CKX53, Japan). The experimental process of the antialgal test is shown in Figure S3.

Antifouling Test in a Marine Environment. For the antifouling test in a marine environment, AFSS and control samples (PVC plate) were immersed in the sea of Dalian, Liaoning, China at the same time (June), and the test samples were periodically checked and recorded for marine organism contamination.

RESULTS AND DISCUSSION

Optimum Parameter of the AFSS. The concentration of different components in the coating solution determines the slippery performance of the AFSS. To obtain the AFSS with good slippery performance, the influence of the concentration of SO, NCC, and capsaicin on the slippery performance of the AFSS was carefully investigated, as shown in Figure 2. The ratio of SO and SCA plays a key role in the construction of the AFSS. When the ratio of SO to SCA was 2:1 (AFSS-SO2), the AFSS was relatively flat and uniform. The surface hardness of the AFSS with different ratios of SO to SCA was tested, and the comparison results are shown in Table 1. It can be seen

Table 1. Surface Hardness of the AFSS

surface types	hardness
AFSS-SO1	3B
AFSS-SO2	3B
AFSS-SO3	4B
AFSS-SO4	5B
AFSS-NCC0.5	3B
AFSS-NCC1	3B
AFSS-NCC2	5B

that the AFSS-SO1 and AFSS-SO2 had the strongest hardness (3B), which is attributed to the high proportion of SCA that could enhance the crosslinking effect. The surface adhesion of the AFSS with different ratios of SO to SCA was also tested, and the comparison results are shown in Table S2. The AFSS-SO2 and AFSS-SO4 exhibited better adhesion (ISO 0). As shown in Figure 2a, the AFSS-SO2 had the minimum initial SA of 2° and the CA was 106°. After immersion in water for 30 days, the SA of AFSS-SO2 increased slightly and remained stable at 15°. The results demonstrated that the AFSS-SO2 had better durability than other surfaces because of its suitable silicone oil content embedded in the coating (Figure 2b). A suitable amount of silicone oil embedded in the coating endowed the AFSS with good slippery performance. In addition, there was no excess free silicone oil on the AFSS. Thus, the suitable ratio of SO to SCA was 2:1.

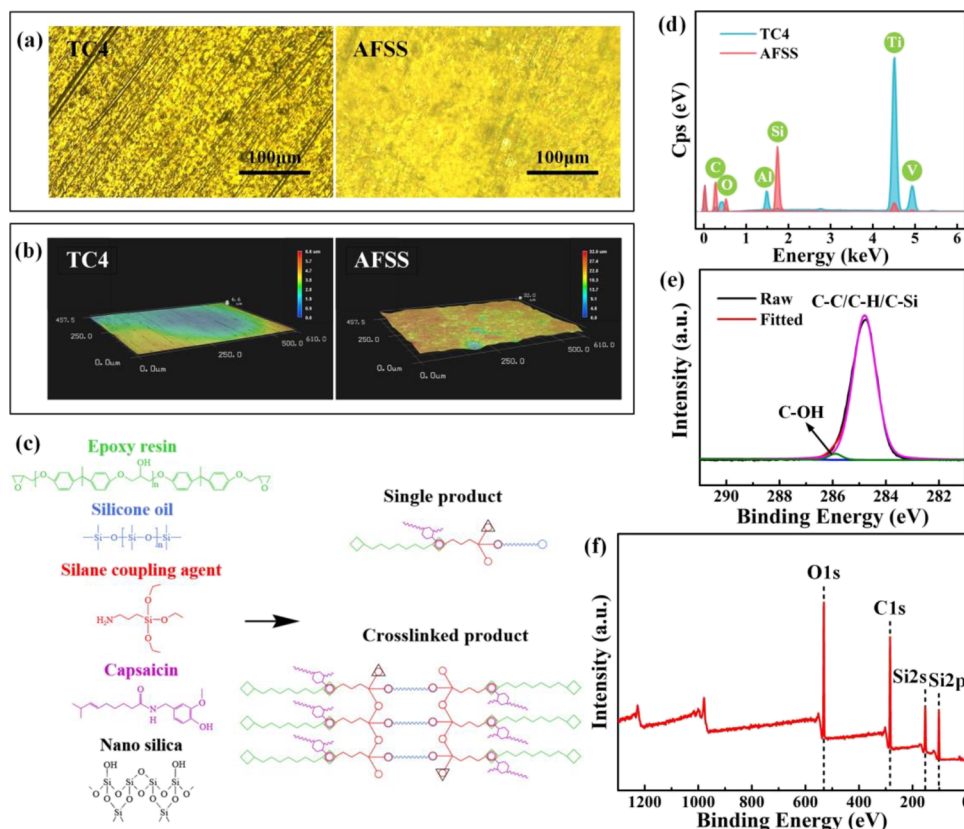


Figure 3. Microscopic morphology (a), three-dimensional morphology (b) of TC4 and the AFSS; schematic of the synthesis processes of the AFSS (c); EDS mapping spectra of the TC4 and AFSS (d); peak-fitted high-resolution C1s spectra of the AFSS (e); XPS spectra of the AFSS (f).

The introduction of NCC could enhance the mechanical strength of the slippery surface. When the ratio of NCC to SO was 1:1 (AFSS-NCC1), the surface was relatively flat and uniform. Similarly, the surface hardness and adhesion of the AFSS with different ratios of NCC to SO were also tested. The results demonstrated that the AFSS-NCC0.5 and AFSS-NCC1 had the strongest hardness (3B) and adhesion (ISO 0). As shown in Figure 2c, AFSS-NCC1 had the minimum SA of 2° and the CA was 106°. After immersion in water for 30 days, the SA increased slightly and remained stable at 15° (Figure 2d). The suitable amount of the NCC endows the AFSS with better mechanical strength and surface stability. Thus, the suitable ratio of NCC to SO was 2:1.

To prevent the growth and reproduction of the bacteria and algae, the nontoxic and environmentally friendly capsaicin was mixed into the coating. The influence of the proportion of capsaicin on the slippery performance of the AFSS was tested. As shown in Figure 2e, the CA of AFSS decreased gradually with the increasing proportion of capsaicin, which is attributed to the hydrophilicity of the capsaicin. Meanwhile, the SA decreased first and then increased with the increasing proportion of capsaicin proportion. When the proportion of capsaicin was 1.5% (AFSS-Cap.1.5%), the AFSS had good slippery performance and the SA was 1.83°. In addition, we also tested the durability of the AFSS-Cap.1.5% after immersion in water. After immersion in water for 30 days, the SA of AFSS-Cap.1.5% increased gradually and stabilized at 12°. The results demonstrated that the durability of AFSS-Cap.1.5% was better than that of the AFSS-Cap.0% (Figure 2f).

Thus, the optimal composition ratio for the preparation of the AFSS was NCC: SO: SCA = 2:2:1 and 1.5 wt % capsaicin.

Synthesis Mechanism. The TC4 titanium alloy surface and AFSS were observed. As shown in Figure 3a, the coating coated on the TC4 titanium alloy substrate was evenly distributed and without obvious grinding. The microstructure consists of nanoscale-size nanoceramic coatings and nanosilica, thus forming a relatively flat slippery surface. In addition, the coating thickness of the AFSS was ~20 μm (Figure 3b). The synthesis process of the AFSS is shown in Figure 3c. The dehydrated SCA condensed with SO to form a cross-linking network, combined with epoxy resin to achieve curing on the substrate, and grafted with nanoparticles to enhance the mechanical strength of the surface. With the aid of capsaicin, the AFSS can resist proteins, bacteria, and algae. Compared with the TC4 titanium sample, the new elements carbon (C), oxygen (O), and silicon (Si) were detected on AFSS as shown in the EDS spectra (Figure 3d). Peak-fitted high-resolution C1s spectra and XPS spectra were then recorded using XPS peak V4.1 software, as shown in Figure 3e,f. The peaks at 284.8 and 286.1 ± 0.1 eV were attributed to C–H/C–C/C–Si and C–OH (Figure 3e). The comparison of FTIR spectra between AFSS and silicone oil is shown in Figure S4. The reaction of each component of the solution was relatively complete, and the AFSS was formed.

Characterization of Surface Energy. The surface energy γ_s of the solid and its dispersion component γ_s^d and polarity component γ_s^p was calculated by measuring the contact angles of the different polarities of the detection liquids on the surface using the Owens–Wendt–Kaelble model equation.³⁷

$$\gamma_s = \gamma_s^p + \gamma_s^d \quad (1)$$

$$\gamma_l(1 + \cos \theta) = 2\sqrt{\gamma_s^p \gamma_l^p} + 2\sqrt{\gamma_s^d \gamma_l^d} \quad (2)$$

where γ_l , γ_l^d , and γ_l^p represent the surface energy, the dispersion force component, and the polarity force component of the detection liquids, respectively. Deionized water was chosen as the polar detection fluid and *n*-hexadecane as the nonpolar detection fluid. The parameters for deionized water and *n*-hexadecane are shown in Table 2.

Table 2. Surface Energy γ_l , Dispersive Force Component γ_l^d , and Polar Force Component γ_l^p of the Detection Liquids

detection liquid	γ_l (mJ/m ²)	γ_l^d (mJ/m ²)	γ_l^p (mJ/m ²)
deionized water	72.8	21.8	51
<i>n</i> -hexadecane	27.5	27.5	0

The surface contact angles θ for deionized water and *n*-hexadecane were measured to be 109.04° and 28.15°, respectively. According to eqs 1 and 2), we calculated γ_s , γ_s^d , and γ_s^p to be 24.3868, 24.34336, and 0.04346 mJ/m², respectively. The calculated results showed that the surface energy of the prepared surface was low. Thus, the adhesion between the contaminant and the AFSS was weak, and the pollutant can be easily removed.

Slippery Performance and Self-Cleaning Performance of the AFSS. *Slippery Performance.* To test the slippery performance of the AFSS, the sliding velocities of water droplets with different volumes were measured. As shown in Figure 4a, water droplets could not slide off and be pinned tightly on the TC4 surface, while the water droplets could easily slide from the AFSS and the sliding velocity of the water droplets increased with the increased volume of the water droplets. In addition, various droplets with different viscosity could also slide off the AFSS, as shown in Figure 4b. The sliding velocities were 2.00, 0.90, 0.89, and 0.17 cm/s for deionized water, coffee, milk, and oil, respectively, demonstrating that the AFSS could repel not only low-viscosity but also high-viscosity liquids. To test the stability of the AFSS, the SA and mass of the AFSS after immersion in water were tested, as shown in Figure 4c. The SA of the AFSS changed little and is still below 10°. Moreover, the mass of the sample had no significant change, indicating no loss of oil in the AFSS. The experimental results demonstrate that the AFSS possesses good durability after immersion in water.

Self-Cleaning Performance. Self-cleaning performance is one of the important indicators of slippery surfaces. The hydrophilic copper chloride dihydrate particles were dissolved in the water droplets dropped on the AFSS and were carried away from the AFSS, demonstrating good self-cleaning performance. By comparison, deionized water dripped onto the surface of the TC4 titanium sample also dissolved the hydrophilic copper chloride dihydrate particles but left water stains and contaminated the whole surface (Figure 4e₁ and Video 1). In addition, the AFSS also showed good self-cleaning performance for hydrophobic contaminants. When the water droplets dropped on the AFSS, the hydrophobic silica particles floated on the water droplets and were carried away from the AFSS. However, the hydrophobic silica particles could not be carried away on the TC4 titanium alloy (Figure 4e₂ and Video 2). Thus, our AFSS shows good self-cleaning performance for both hydrophilic and hydrophobic contaminants.

Antifouling Performance. To test the antifouling performance of the AFSS, the TC4 titanium alloy plate was set as the control group, and the capsaicin surface (CS) and slippery surface (SS) were set to verify the synergistic effect of the slippery surface and capsaicin.

Antiprotein Performance. Protein is the main component of the first stage of marine fouling, which provides a nutrient layer for bacteria and other microorganisms. The antiprotein adhesion property was tested by BSA adsorption (Figure 5a–c). According to the fitting standard curve and calculation, the BSA adsorption of the TC4 titanium alloy plate was ~0.0212 mg/cm². The BSA adsorption of CS was ~0.0148 mg/cm², which indicated that the capsaicin could resist the adhesion of protein. The BSA adsorption of SS was ~0.0075 mg/cm², which is attributed to the low adhesion between the protein and the slippery surface. The BSA adsorption of AFSS was only 0.0062 mg/cm², which is attributed to the synergistic effect of the capsaicin and the slippery performance. The results indicated that the AFSS is effective in preventing protein adhesion.

Antibacterial Performance. An important stage of marine fouling is the formation of biofilm, and bacteria are one of the main components of the formed biofilm. The antimicrobial adhesion property against Gram-negative bacteria (*E. coli*) of the AFSS was tested, as shown in Figure 5d. It can be seen that the dense colonies of the medium photocopied from the TC4 titanium alloy plate were formed and the density of the colonies was ~3.40/cm² (medium diameter 9 cm). By comparison, only a few colonies were observed from the medium photocopied from the CS and SS and the density of the colonies for the CS and SS was 1.35 and 0.13/cm², respectively, which demonstrated that both the CS and SS could resist the growth of the bacteria. For the AFSS integrated with the slippery performance and the capsaicin, the density of the colonies was the minimum and was ~0.09/cm², which was 97.2% lower than that of TC4. The antimicrobial adhesion property against Gram-positive bacteria (*S. aureus*) of the AFSS was tested, as shown in Figure 5e. Similar to *E. coli*, the dense colonies of the medium photocopied from the TC4 titanium alloy plate were formed and the density of the colonies was ~0.25/cm². AFSS had very few colonies on the surface, and the density was 0.02/cm², indicating a good antibacterial adhesion property of the AFSS.

Antialgal Performance. Algae are also one of the main components of marine fouling. We further investigated the antialgal adhesion performance of the AFSS, as shown in Figure 5e. After incubation in the algal solution for 7 days, a considerable amount of algae covered the surface of the untreated TC4 titanium alloy plate, and the algal adhesion ratio was 46.94%. By comparison, the algal adhesion ratio on CS and SS was 4.58 and 4.12% respectively, while no algae was observed on AFSS because of the synergistic effect of the slippery performance and the capsaicin could resist the growth of the algae (Figure 5c). The experimental results indicated that the AFSS had good antialgal adhesion performance.

Antifouling Performance in the Marine Environment. To verify whether the developed AFSS can be applied in a realistic marine environment, we investigated the antifouling performance of the AFSS in a marine environment, as shown in Figure 6. The PVC plate as the control group showed obvious yellow and black stain adhesion after 15 days of immersion in the real sea. When the experiment progressed to 30 days, a large number of marine fouling organisms attached to the surface

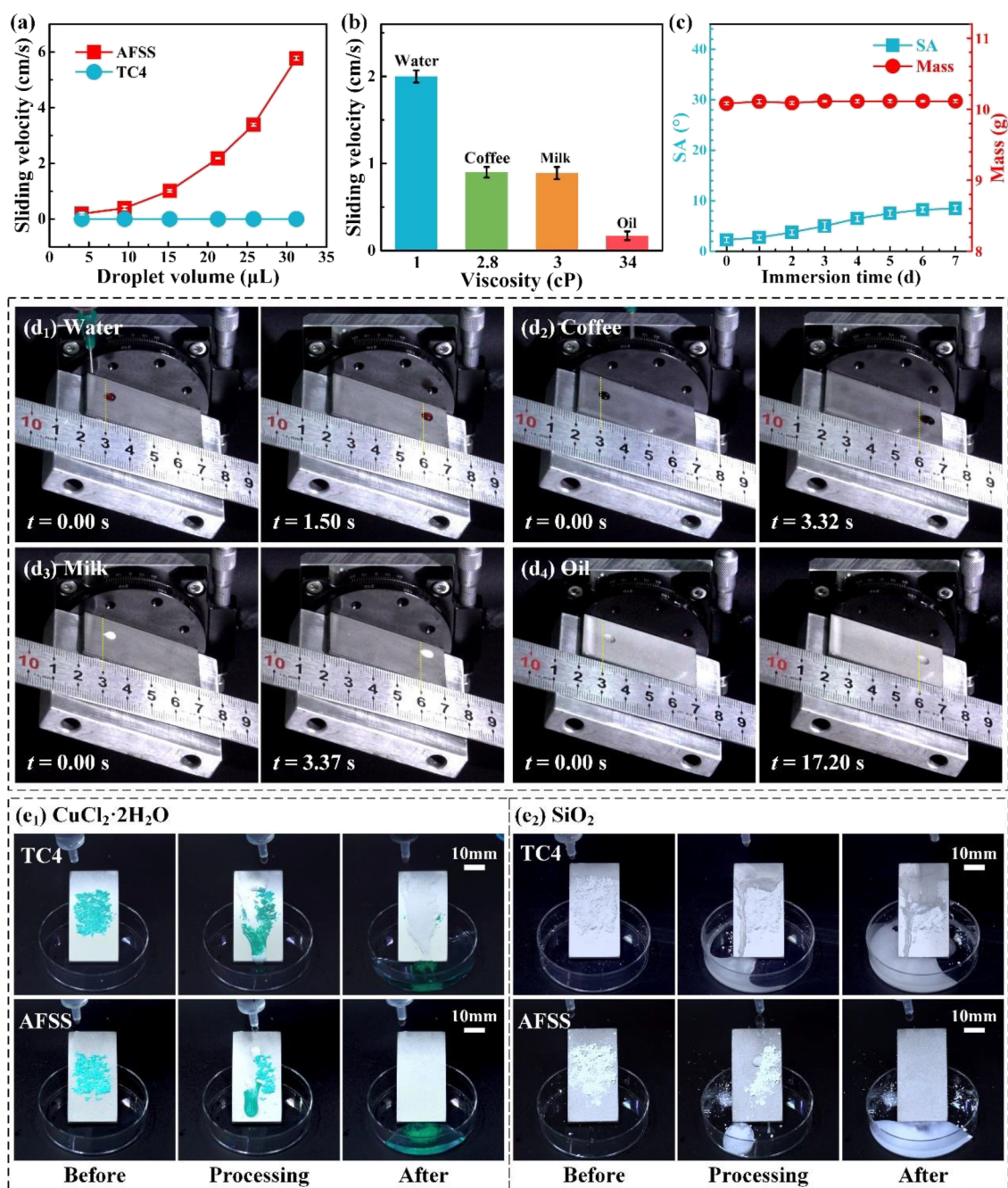


Figure 4. Slippery performance and self-cleaning performance of the AFSS. (a) Variation of sliding velocities on the AFSS for water droplets with different volumes. (b) Variation of sliding velocities of droplets with different viscosity on the AFSS. (c) Variation of water SA on the AFSS and mass of the samples with the immersion time. (d) Images of various droplets (water, coffee, milk, and oil) sliding processes on the AFSS; images show the self-cleaning processes of the contaminants on the TC4 and AFSS. (e₁) Hydrophilic copper chloride dihydrate particles and (e₂) hydrophobic silica particles.

grew. The surface of the AFSS was still clean and smooth during the immersion process in the same marine environment, and no fouling organisms observed on the AFSS, which showed good antifouling performance and durability in the actual marine environment.

CONCLUSIONS

In summary, we developed an antifouling slippery surface with good slippery performance and marine antifouling properties. The AFSS could be easily obtained by immersing TC4 into the coating solution composed of silicone oil, silane coupling

agent, nanosilica, nanoceramic coating, capsaicin, and epoxy resin. The as-prepared surface possessed excellent slippery performance with an SA of $\sim 1.8^\circ$ and good durability with a stable SA of 12° after 30 days of immersion in water. Compared with the original TC4, the AFSS exhibited excellent slippery performance for various droplets and good self-cleaning performance. In addition, the AFSS had superior antifouling performance, and the adhesion of proteins, bacteria, and algae was reduced by 70.7%, 97.2%, and 97.7% respectively. The AFSS also showed good antifouling ability and durability in the actual marine environment. Therefore,

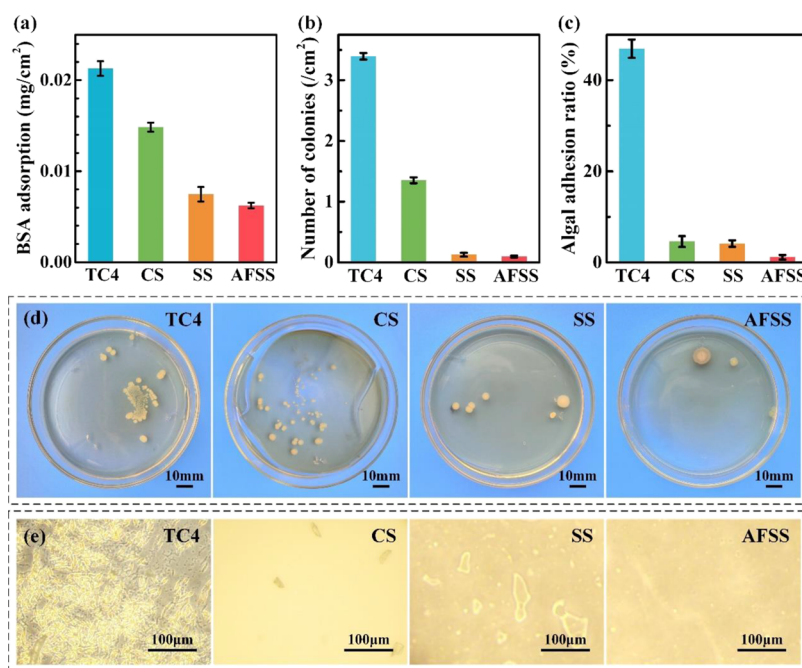


Figure 5. Antifouling performance of the AFSS. (a) BSA adsorption of the TC4, CS, SS, and AFSS after immersion in the protein solution for 12 h. (b) Number of the *E. coli* colonies of the medium photocopied from the TC4, CS, SS, and AFSS after incubation for 24 h. (c) Algal adhesion ratio of on the TC4, CS, SS, and AFSS after incubation in the algal solution for 24 h. (d) Digital photos of the *E. coli* colonies of the medium photocopied from the TC4, CS, SS, and AFSS after incubation for 24 h. (e) Fluorescence microscope photos on the TC4, CS, SS, and AFSS after incubation in the algal solution for 24 h.

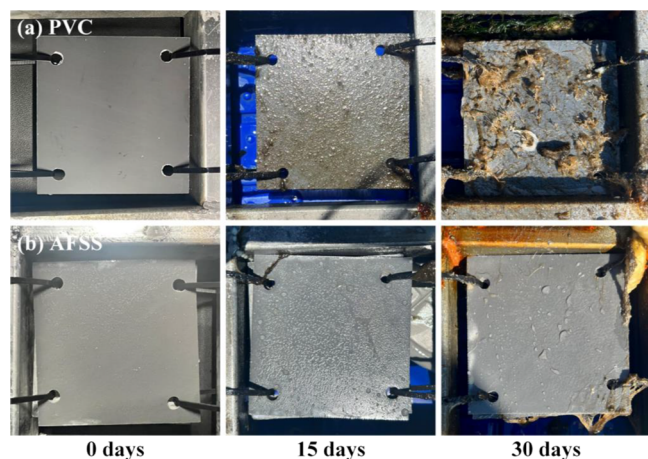


Figure 6. Antifouling performance of the AFSS in the marine environment. (a) Digital photos of the PVC panels in marine environments within 0 to 30 days; (b) digital photos of the AFSS in marine environments within 0 to 30 days.

this AFSS could be developed as a promising antifouling coating in titanium alloy piping systems in marine engineering.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.langmuir.3c00986>.

Self-cleaning performance of the AFSS for hydrophilic contaminants (MP4)

Self-cleaning performance of the AFSS for hydrophobic contaminants (MP4)

Standard protein curve, experimental process of the antibacterial test, experimental process of the antialgal test,

FTIR spectra of AFSS and silicone oil, antibacterial performance of AFSS, properties of common oils, and surface adhesion of the AFSS (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Jinlong Song — State Key Laboratory of High-Performance Precision Manufacturing, Dalian University of Technology, Dalian 116024, P. R. China; orcid.org/0000-0002-3769-2069; Email: songjinlong@dlut.edu.cn

Authors

Xin Liu — State Key Laboratory of High-Performance Precision Manufacturing, Dalian University of Technology, Dalian 116024, P. R. China; orcid.org/0000-0001-8323-7289

Xiaolei Gu — State Key Laboratory of High-Performance Precision Manufacturing, Dalian University of Technology, Dalian 116024, P. R. China

Yuyang Zhou — State Key Laboratory of High-Performance Precision Manufacturing, Dalian University of Technology, Dalian 116024, P. R. China; orcid.org/0000-0002-4806-6266

Weihao Pan — State Key Laboratory of High-Performance Precision Manufacturing, Dalian University of Technology, Dalian 116024, P. R. China

Jiyu Liu — State Key Laboratory of High-Performance Precision Manufacturing, Dalian University of Technology, Dalian 116024, P. R. China; orcid.org/0000-0003-3313-5721

Complete contact information is available at:

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Notes

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